SIEM :

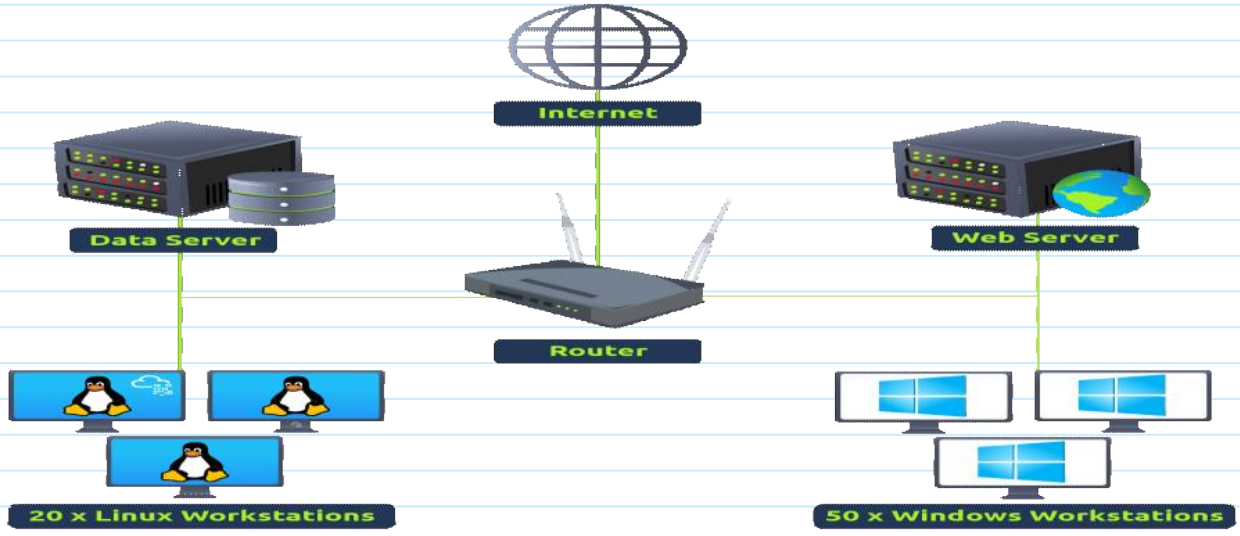
Saturday, January 10, 2026 3:49 PM

Introduction to SIEM :

Security Information and Event Management system (SIEM) is the core security solution that a SOC analyst uses in the security operations center. we will learn how different devices in a network generate logs and why it's essential to have a centralized solution to collect, normalize, and correlate these logs.

Logs Everywhere :

In any network, multiple devices constantly communicate with each other and, most of the time, with the Internet through a router. These devices include workstations, servers, and network infrastructure components.



Log sources can be broadly divided into two main categories, discussed below.

1) Host-Centric Log Sources :

- **Host-centric log sources capture events that occur on or directly related to an individual system. These logs provide visibility into what is happening inside endpoints and servers.**
- **Devices that generate host-centric logs include:**
  - **Windows systems**
  - **Linux systems**
  - **Application and database servers**

Examples of Host-Centric Logs :

- **A user accessing or modifying a file**
- **A user attempting to authenticate**
- **Process creation or execution events**
- **Registry key or value creation, modification, or deletion**
- **PowerShell execution activity**

These logs are critical for understanding user behaviour and process activity on a specific machine.

2) Network-Centric Log Sources :

- ★ **Network-centric log sources capture events generated when systems communicate with each other or access external resources over the network.**
- ★ **Devices that generate network-centric logs include:**
  - **Firewalls**
  - **IDS/IPS**
  - **Routers**
  - **VPN gateways**

Examples of Network-Centric Logs :

- **SSH connections**
- **File transfers over FTP**
- **Web traffic (HTTP/HTTPS)**
- **Remote access through VPN**
- **Network file-sharing activity**

These logs provide visibility into traffic flow, connections, and data movement across the network.

Challenges of Analyzing Logs Manually :

★ Numerous Log Sources :

Modern networks contain a large number of log sources, each generating hundreds or even thousands of events per second.

★ No Centralisation :

Because logs reside on the systems that generate them, analysts often need to manually connect to each device using tools such as

★ Limited Context :

A single log entry rarely tells the whole story. Many activities look harmless when viewed in isolation.

Why SIEM?

In the previous task, we explored how different log sources generate massive volume of logs in many formats—and how difficult it is to analyze them effectively.

- ★ **This is where Security Information and Event Management (SIEM) comes into play.**
- ★ **A SIEM solution collects logs from a wide range of log sources, standardizes them into a common format, correlates related events, and detects malicious activity using predefined and custom detection rules.**

Core Features of a SIEM Solution :

- ★ **Centralized Log Collection :**

SIEM solutions collect logs from all types of log sources, including:

- **Endpoints**
- **Servers**
- **Firewalls**
- **IDS/IPS**
- **Cloud platforms**

Logs are typically ingested using lightweight agents, APIs, or log forwarding mechanisms and stored in a central location.

★ Normalization of Logs :

Raw logs differ greatly in structure and format:

- **Windows Event Logs look different from Linux logs**
- **Firewall logs differ from application logs**

Because SIEM centralizes logs, it also:

- **Parses logs by breaking them into structured fields (e.g., user, IP, action)**
- **Normalizes logs by converting them into a consistent format**

★ Correlation of Logs :

On their own, individual log entries often provide limited value. SIEM shines by correlating events across multiple sources to uncover suspicious patterns.

Example Correlation Scenario :

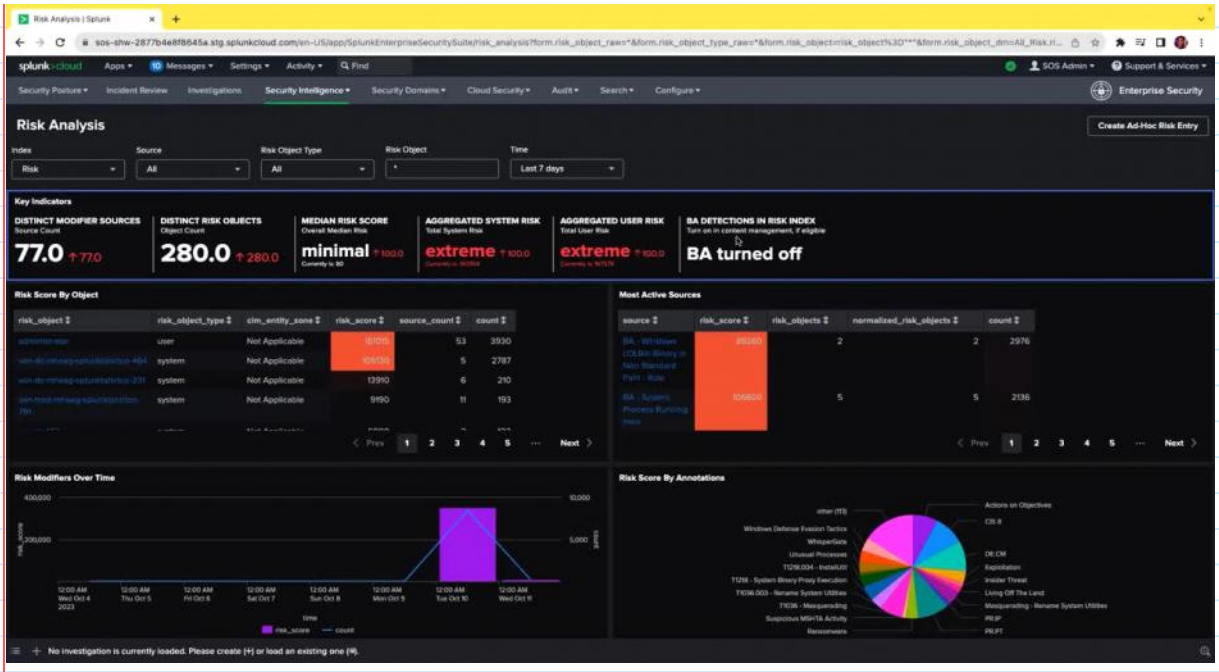
- **Haris logs in via VPN from a previously unseen IP address**
- **Haris accesses documents on a shared drive**
- **Haris executes a PowerShell script**
- **The system makes an outbound network connection**

★ Real-Time Alerting :

SIEM platforms detect malicious activity using detection rules.

- **Many rules are provided out of the box**
- **Analysts can create custom rules to improve detection over time**
- **When rule conditions are met, alerts are generated in real time**

Analysts are immediately notified and can investigate alerts directly within the SIEM interface.



Dashboards are one of the most powerful components of any SIEM. They present complex data in a clear, visual, and actionable format.
After logs are ingested and normalized, SIEM dashboards provide insights such as:

- **Alert highlights**
- **System notifications**
- **Health alerts**
- **Failed login attempts**
- **Event ingestion counts**
- **Rules triggered**
- **Top domains visited**

Most SIEM solutions include default dashboards and also support custom dashboard creation tailored to SOC requirements.

Log Sources and Ingestion :

Every device in a network generates logs whenever an activity occurs. Examples include:

★ Windows Machine :

Windows systems record nearly every activity in the form of Windows Event Logs. Each event type is assigned a unique Event ID, which helps analysts quickly identify and track specific actions.
To view logs on a Windows system:

- **Open the Start menu**
- **Type Event Viewer**
- **Navigate through log categories such as Security, System, and Application**

★ Linux Machine

Linux systems store logs as text files, typically under the /var/log directory. These logs capture system events, errors, warnings, and application activity and are continuously ingested into the SIEM.

Common Linux Log Locations :

- /var/log/httpd – HTTP request, response, and error logs
- /var/log/cron – Cron job execution events
- /var/log/auth.log or /var/log/secure – Authentication-related logs
- /var/log/kern.log – Kernel-related events

★ Web Server Logs

Web servers are critical log sources because they record all inbound and outbound web traffic, making them essential for detecting web-based attacks.
On Linux-based systems, Apache logs are commonly stored in:

- /var/log/apache
- /var/log/httpd

Log Ingestion into SIEM

All these logs provide a wealth of security-relevant information. However, they only become truly useful once they are ingested into a SIEM. Each SIEM platform supports multiple ingestion methods.

Common Log Ingestion Methods :

Agent / Forwarder:

SIEM vendors provide lightweight agents (called forwarders in some platforms) that are installed on endpoints. These agents collect logs and securely forward them to the SIEM server.

Syslog :

Syslog is a widely used protocol for sending real-time logs from systems such as:

- Linux servers
- Network devices
- Web servers

Logs are forwarded to a centralized SIEM destination for analysis.

Manual Upload :

Some SIEM platforms allow analysts to upload offline log files for investigation.
This is useful during:

- Incident response
- Log review from compromised systems
- Historical analysis

Once ingested, logs are parsed, normalized, and made searchable.

Port Forwarding / Listener-Based Ingestion :

SIEM solutions can be configured to listen on specific ports, allowing endpoints and devices to forward logs directly to the SIEM over the network.

Alerting Process and Analysis :

We’ ve learned that a SIEM detects threats by correlating logs from multiple log sources and triggering alerts.
But what actually causes an alert to fire?

The answer lies in detection rules.
A SIEM uses detection rules to identify suspicious or malicious behaviour. These rules are critical for timely threat detection, allowing analysts to respond before damage escalates.

What Are Detection Rules :

Detection rules are essentially logical expressions that define what to look for and when to raise an alert.
They continuously evaluate incoming logs and trigger when specific conditions are met.

Simple Examples of Detection Rules :

- **If a user has 5 failed login attempts within 10 seconds, trigger an alert for Multiple Failed Login Attempts**
- **If a successful login occurs after multiple failures, trigger an alert for Brute Force Success**
- **Trigger an alert whenever a USB device is inserted (useful if USB usage is restricted)**
- **If outbound data transfer exceeds 25 MB, raise an alert for Potential Data Exfiltration**
  (Thresholds usually depend on company policy)